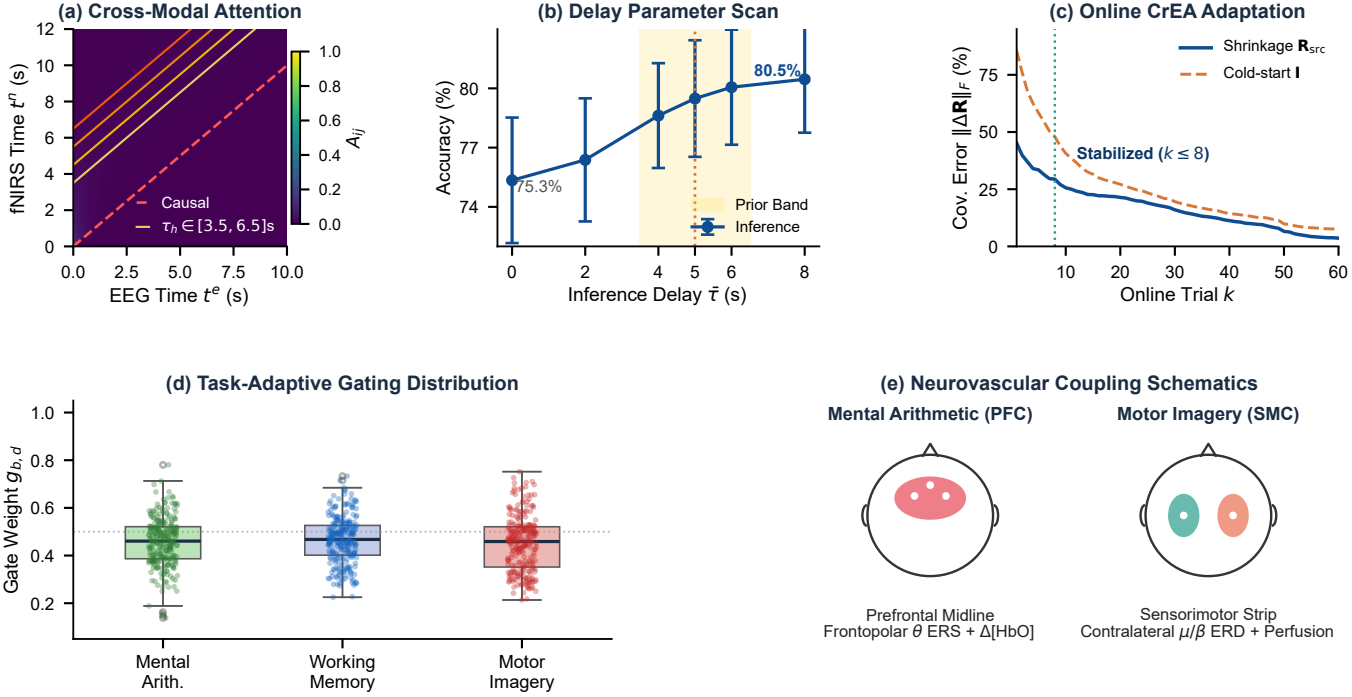

(a) Cross-Modal Attention
fNIRS Time t^n (s)
EEG Time t^e (s)
Causal
$\tau_h \in [3.5, 6.5]$s
A_{ij}
(b) Delay Parameter Scan
Accuracy (%)
Inference Delay $\bar{\tau}$ (s)
75.3%
80.5%
Prior Band
Inference
(c) Online CrEA Adaptation
Cov. Error $\|\Delta\mathbf{R}\|_F$ (%)
Online Trial k
Shrinkage $\mathbf{R}_{\text{src}}$
Cold-start $\mathbf{I}$
Stabilized ($k \leq 8$)
(d) Task-Adaptive Gating Distribution
Gate Weight $g_{b,d}$
Mental Arith.
Working Memory
Motor Imagery
(e) Neurovascular Coupling Schematics
Mental Arithmetic (PFC)
Motor Imagery (SMC)
Prefrontal Midline
Frontopolar θ ERS + Δ[HbO]
Sensorimotor Strip
Contralateral μ/β ERD + Perfusion